EEE3096S Practical 3 — Voltage Mirror

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Date: 23 October 2026 Build configuration: -O0

1. Measurements and predictions

Measured analogue reference V_R: 3.2946 V

DMM range/resolution: 10 V DC range, 5 significant figures (0.1 mV resolution)

Nominal ADC/DAC interval $V_R/4096$: $3.2946/4096 = 0.8043$ mV

Predicted coarse output step $V_R/64$: $3.2946/64 = 51.48$ mV

Input target	Mode	Measured PA5 (V)	C	D	Predicted PA4 (V)	Measured PA4 (V)
∼0.800 V	FULL12	0.79955	992	992	0.7979	0.80163
∼0.800 V	COARSE6	0.79955	992	960	0.7722	0.77765
∼1.65 V	FULL12	1.6528	2045	2045	1.6449	1.6536
∼1.65 V	COARSE6	1.6528	2053	2048	1.6473	1.6536
∼2.500 V	FULL12	2.5040	3113	3113	2.5039	2.4753
∼2.500 V	COARSE6	2.5040	3116	3072	2.4710	2.4753

Generator: 1.996 Hz; 1.77 Vpp; 1.787 V offset

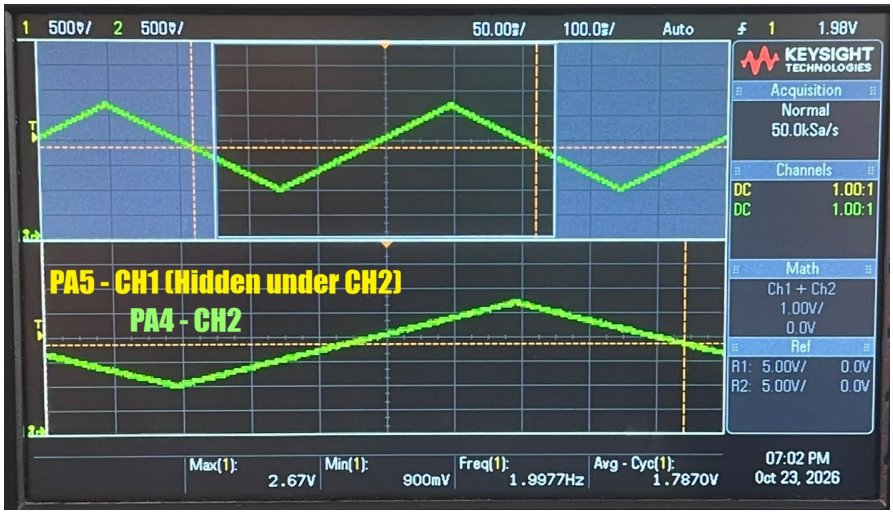
Verified PA5 minimum/maximum: 0.9 / 2.67 V

Measured adjacent coarse step: 51.25 mV (average over six steps)

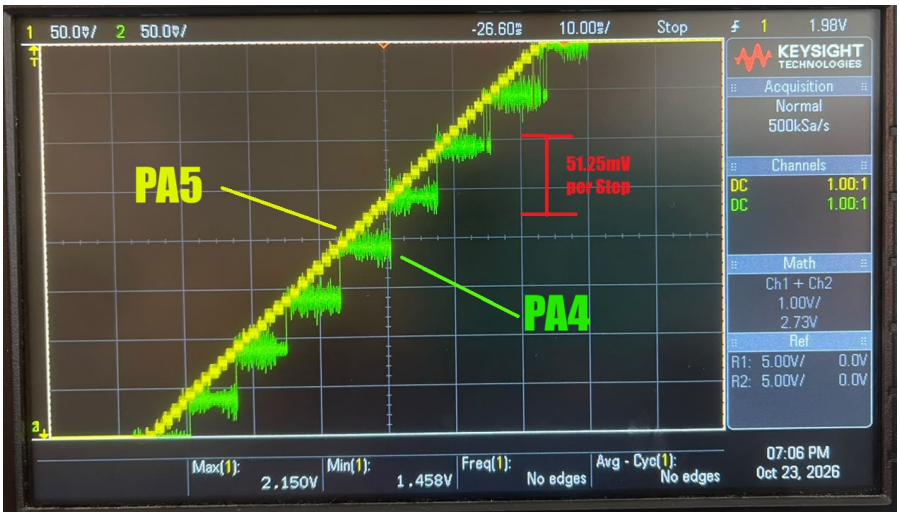
Measured minus predicted step: $51.25 - 51.48 = -0.23$ mV

Percentage difference $100(\Delta V_{\text{measured}} - \Delta V_{\text{predicted}})/\Delta V_{\text{predicted}}$: -0.44 %

2. Scope captures



Capture 1 — FULL12. CH1 = PA5 input, CH2 = PA4 output, both 500 mV/div, DC coupled; 50 ms/div (zoomed window of the 100 ms/div main timebase). The PA4 trace lies on top of PA5 over the whole cycle (PA5: 0.90–2.67 V, 1.998 Hz).



Capture 2 — COARSE6. CH1 = PA5 input (yellow), CH2 = PA4 output (green), both 50 mV/div, DC coupled; 10 ms/div, rising ramp. PA4 is a staircase of plateaus below the input ramp; the average spacing of adjacent settled plateaus over six steps is 51.25 mV.

3. Explanation

Mapping. FULL12 copies the full 12-bit code, $D = C$, so all 4096 commands are available. COARSE6 limits resolution by clearing the six least significant bits, $D = C \,\&\, \texttt{0b111111000000} = 64\lfloor C/64 \rfloor$ This gives us $4096/64 = 64$ distinct commands (0, 64, ..., 4032) spaced 64 codes apart, or $V_R/64 = 51.48$ mV at the output. The largest is 4032 since the next multiple of 64, 4096, needs a 13th bit. Because bits are cleared, COARSE6 always rounds down ($D \leq C$), so the output sits 0–63 LSB (0–51 mV) below the input, as the staircase in Capture 2 shows. The ADC and DAC both remain 12-bit; only the software mapping removes the six bits, giving an effective 6-bit resolution.

Measured steps. The measured step of 51.25 mV, which we averaged over six adjacent plateaus, is 0.23 mV (−0.44 %) below the prediction 51.48 mV. This difference is within the expected measurement uncertainty, which comes mainly from noise on the plateaus, cursor placement and the scope's limited vertical resolution, so the measured step agrees with $V_R/64$.

Resolution and accuracy. The FULL12 trace in Capture 1 only *looks* smooth. The DAC still outputs discrete codes, updated every 0.5 ms, so PA4 is a staircase of steps of about 3.5 mV on this triangle. These steps are too small to see at 500 mV/div, so a smooth trace shows only that the steps are below the scope's resolution, not that the output is continuous.

Resolution is the smallest output step: 0.80 mV in FULL12 and 51.48 mV in COARSE6. *Accuracy* is how close the output is to the true value, and is set by the converters' offset, gain and linearity errors and by noise, not by the number of bits. Our FULL12 outputs at 0.8 V and 1.65 V are 3.7–8.7 mV or 5–11 LSB from their predicted voltages, an error of about 0.5 %; that is to say, the conversion is fairly accurate.

One DC result (0.8 V, FULL12). With $C = 992$, the ADC estimate $CV_R/4096 = 0.7979$ V is 1.6 mV (≈ 2 LSB) below the measured input 0.7996 V. With $D = C = 992$, the nominal DAC output $DV_R/4096 = 0.7979$ V is 3.7 mV (≈ 5 LSB) below the measured output 0.8016 V, so the output is 2.1 mV above the input. Copying the code is exact only digitally: the ADC has its own offset, gain and linearity error, the DAC has independent errors, and noise makes C vary by a few LSB.